Listing 2.4 Transform a PID document to JSON representation using Claude Sonnet 4

```
import base64, requests

def analyze_image(api_key, image_path, prompt):
  with open(image_path, "rb") as f:
    image_data = base64.b64encode(f.read()).decode('utf-8')

  response = requests.post(
    "https://api.anthropic.com/v1/messages",
    headers={
      "x-api-key": api_key,
      "anthropic-version": "2023-06-01",
      "content-type": "application/json"
    },
    json={
      "model": "claude-sonnet-4-20250514",
      "max_tokens": 5000, #1
      "messages": [{
        "role": "user",
        "content": [
          {"type": "text", "text": prompt},
          {"type": "image", "source": {
            "type": "base64",
            "media_type": "image/png",
            "data": image_data
          }}
        ]
      }]
    }
  )
  return response.json()["content"][0]["text"]

if __name__ == "__main__":
  api_key = "sk_***"

  result = analyze_image(
    api_key,
    "images/pid.png",
    "Convert this diagram to JSON" #2
  )
```

#1 This parameter defines the maximum number of tokens the model should generate before stopping. Customize this to the number of tokens you need.